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RESEARCH ARTICLE

A new cure rate model with flexible competing causes with applications to melanoma and transplantation data

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In this article, we introduce a long-term survival model in which the number of competing causes of the event of interest follows the zero-modified geometric (ZMG) distribution. Such distribution accommodates equidispersion, underdispersion, and overdispersion and captures deflation or inflation of zeros in the number of lesions or initiated cells after the treatment. The ZMG distribution is also an appropriate alternative for modeling clustered samples when the number of competing causes of the event of interest consists of two subpopulations, one containing only zeros (cure proportion), while in the other (noncure proportion) the number of competing causes of the event of interest follows a geometric distribution. The advantage of this assumption is that we can measure the cure proportion in the initiated cells. Furthermore, the proposed model can yield greater or lower cure proportion than that of the geometric distribution when modeling the number of competing causes. In this article, we present some statistical properties of the proposed model and use the maximum likelihood method to estimate

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